Vent PROTOTYPE

Manufacturing intelligence, *built for the floor.*

An AI-driven platform that captures operator knowledge in real time and turns it into structured intelligence for QA, manufacturing management, and facility leadership — in a GMP-compliant, regulation-ready system.

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Dogpatch Labs Application
February 2026

LIVE AT

vent-production-8f8a
.up.railway.app

- ✓ Working prototype
- ✓ Floor-validated
- ✓ 21 CFR Part 11 ready
- ✓ Bootstrapped

THE PROBLEM

The floor knows. The facility doesn't.

In regulated biologics manufacturing, the operators closest to the process are the first to spot deviations, inefficiencies, and procedure gaps. Almost none of it gets captured in time to be useful.

🚨

Reporting is feared

Deviation systems are traceable. Operators know that raising an issue means their name is attached. So observations go unreported, and the facility operates blind.

📋

Documentation consumes the day

30–40% of facility man-hours go to documentation. SOPs live in PDFs nobody reads proactively. Deviations are filed weeks after they happen. The signal is always delayed.

📉

Signal degrades at every handoff

By the time an observation reaches QA or management, it has been filtered, delayed, and summarised. The original insight — the one that could prevent the next deviation — is gone.

> *"The same deviation happens three times across three shifts before anyone connects the dots — because the system isn't designed to connect them."*
> — A reality observed daily on biologics manufacturing floors

THE COST

This costs the industry billions.

These aren't theoretical numbers. They are the operating reality of every regulated biologics facility in the world right now.

40–60
man-hours
per CAPA investigation event
Source: industry benchmark

€1–5M
per batch failure
in biologics drug substance manufacturing
Source: pharma industry estimates

30–40%
of facility man-hours
consumed by documentation in GMP operations
Source: GMP documentation burden studies

60–90
days
average deviation-to-closure cycle
under traditional QMS workflows
Source: pharma QMS benchmarks

💡 And behind every one of these numbers is a human cost: the operator who noticed something and said nothing, the QA reviewer buried in paperwork, the director making decisions on data that's already two weeks old.

THE SOLUTION

Vent is an intelligence amplifier.

It takes the deep, real-time knowledge that exists in the heads and hands of the people closest to the process — and makes that knowledge instantly actionable at every level of the organisation.

WITHOUT VENT

- ✗ Operators swallow observations to avoid blame
- ✗ SOPs sit in binders nobody reads in the moment they're needed
- ✗ QA manually cross-references deviations against SOP libraries
- ✗ CAPA packages built from scratch, 40+ hours each
- ✗ Recurring patterns invisible until a third or fourth occurrence
- ✗ Director gets a weekly summary written by someone else

WITH VENT

- ✓ Operators submit anonymously in 30 seconds, from any device
- ✓ SOP query in plain language — step-by-step answer in seconds
- ✓ AI cross-references every submission against all relevant SOPs automatically
- ✓ Structured CAPA package drafted in seconds, reviewed by QA
- ✓ Pattern detection flags recurring issues on first repeat occurrence
- ✓ Director sees live facility intelligence — not a weekly summary

HOW IT WORKS

Four layers. One closed loop.

1

Capture

OPERATOR

Anonymous 30-second observation submission. Plain language. Photo/video attachments. No form templates, no deviation numbers, no fear of blame.

→ Raise an observation
→ Query a procedure
→ Build a document

2

Analyse

AI ENGINE

Claude + VoyageAI RAG pipeline. Semantic SOP search, root cause hypothesis, priority classification, CAPA generation, contact routing, pattern detection.

→ SOP cross-reference
→ Risk level assessment
→ Pattern matching

3

Workflow

QA & MFG SUPPORT

Structured 4-phase lifecycle. Electronic signatures on approval transitions. Immutable audit trail. SOP change management. 21 CFR Part 11 / EU Annex 11 compliant.

→ QA review & sign-off
→ CAPA tracking
→ Doc control

4

Intelligence

LEADERSHIP

Live facility dashboard. Submission trends by area, priority, and status. CAPA metrics. Cross-submission pattern analysis. Real decisions, not weekly summaries.

→ Director dashboard
→ Trend analysis
→ Compliance reporting

Every submission tightens the loop. Every observation makes the facility smarter.

PRODUCT — OPERATOR LAYER

Designed for the floor, not the office.

Two core tools, built for people whose hands are busy. No training required. No system navigation. Just ask or report.

vent-production.up.railway.app/query

Vent • Query SOPs Doc Builder Operator

Ask the *procedures.*

Search the facility SOPs in plain language. Get step-by-step instructions and exact parameter values.

pH control | Inoculation criteria | DO control cascade | Bleed rate

Ask a question about any SOP procedure, parameter, or step...

— Raise an observation

Raise an observation

Anonymous · routed to QA & MSAT

OBSERVATION

What did you observe? Any deviation, concern, or SOP gap?

Upstream Bioreactor ▾ | Low | Med | High

🔒 Anonymous by design. Your identity is never recorded.

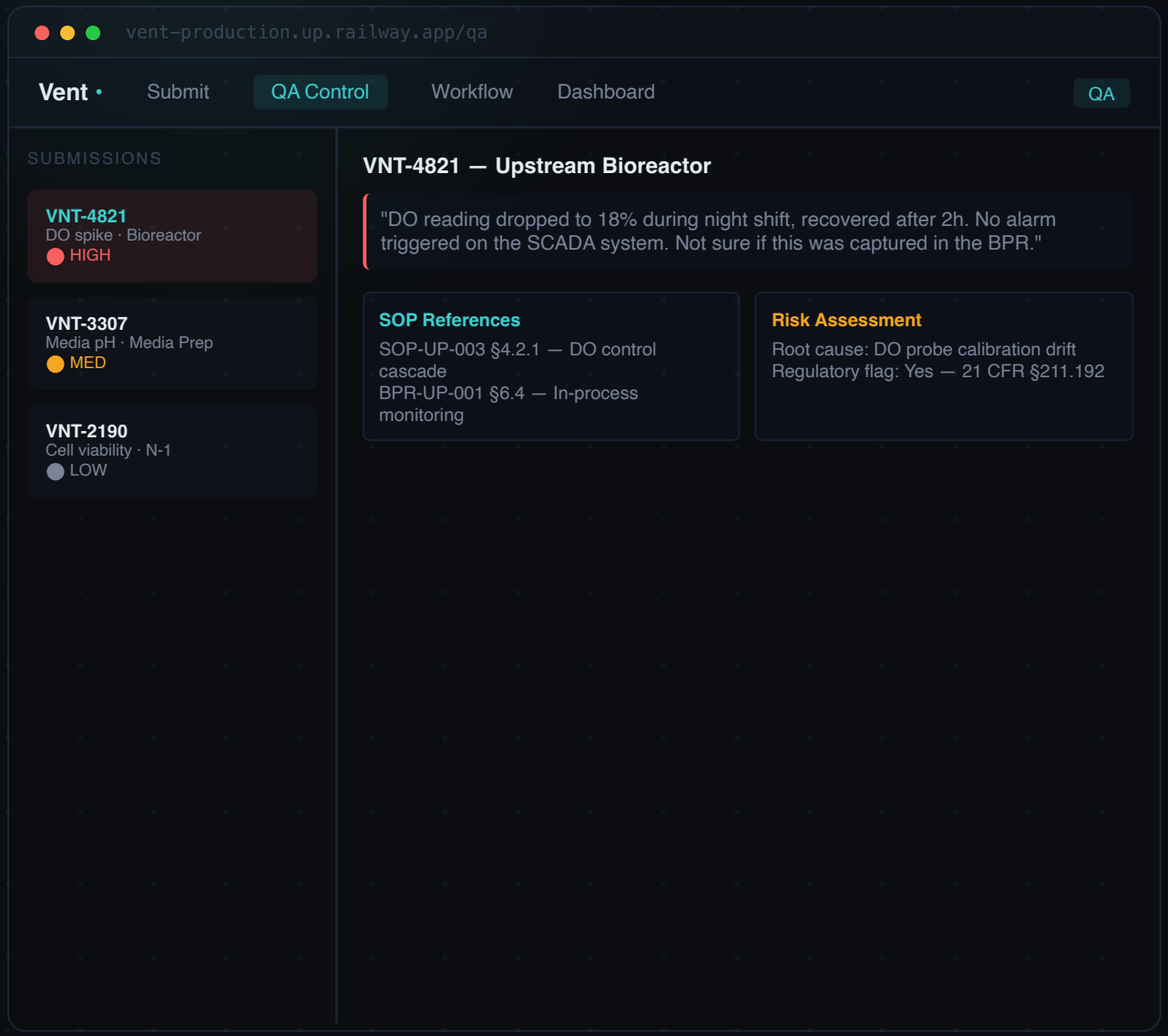
PHOTOS OR VIDEO — OPTIONAL

Add photos or video
Drag & drop or click · up to 5 files

Submit observation →

QA has a command centre, not a queue.

Every submission arrives fully analysed — SOP references, risk level, corrective actions, and contact routing already generated. QA reviews, decides, signs.



Auto SOP cross-reference

Every submission arrives with the exact SOP sections cited by ref number. Click any reference to read the full section inline.

CAPA package drafted

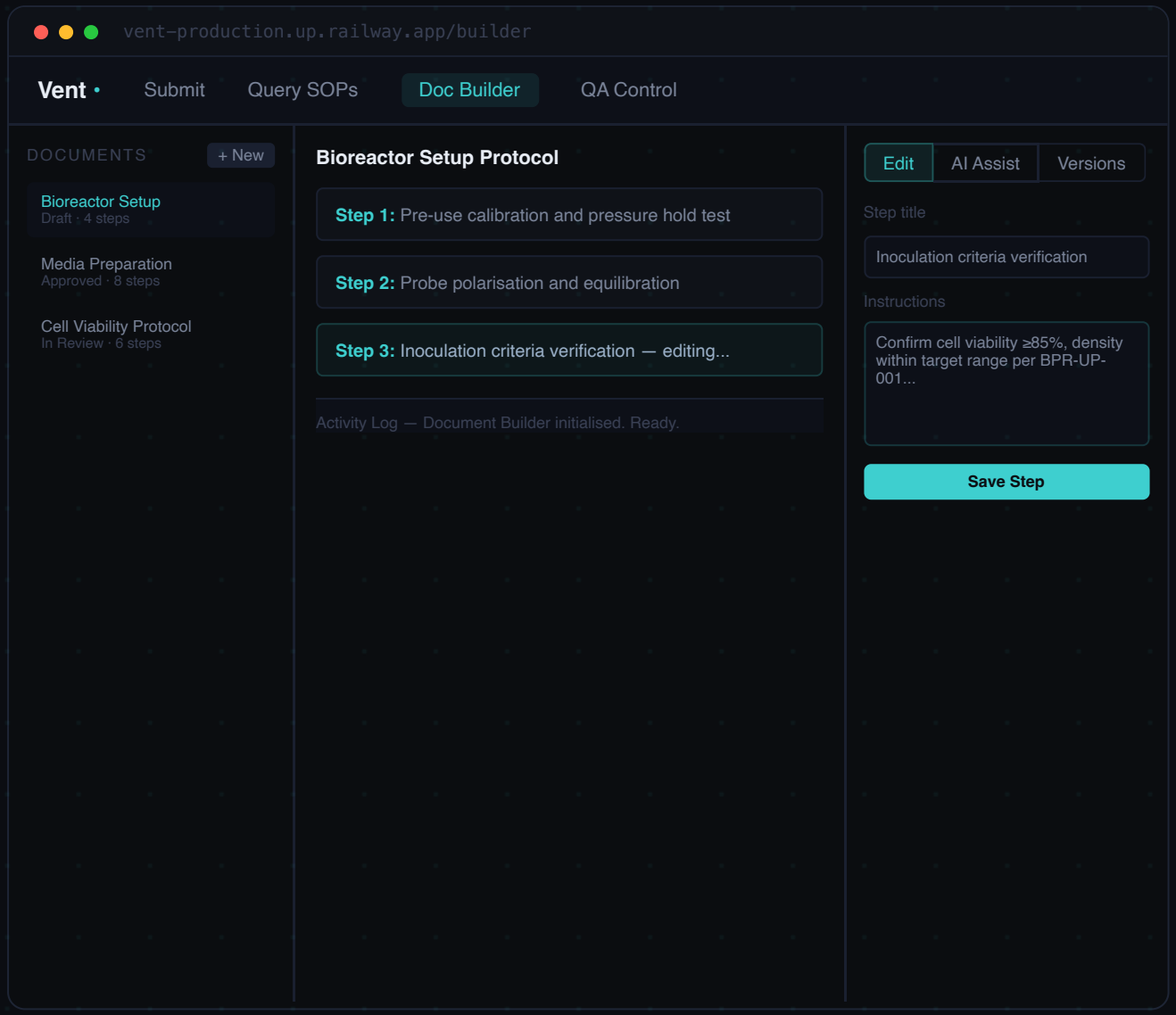
Corrective actions generated with timing bands — immediate, short-term, long-term — and automatically assigned to the right contact.

Electronic signatures

QA approval, director sign-off, and closure require user ID + reason — logged with timestamps and SHA-256 checksums.

Living documents, authored by the people who use them.

A full IDE-style authoring environment for SOPs, work instructions, and batch record templates — built into the same platform operators use every day.



Key capabilities

- ✓ Step-based document structure with version snapshots
- ✓ Drag-to-reorder steps — VS Code-style IDE layout
- ✓ Draft → In Review → Approved lifecycle tracking
- ✓ AI Assist tab — Claude-powered draft generation
- ✓ Real-time activity log for audit compliance
- ✓ Media attachments — photos, video, files

Operators author documents in the same tool they use for queries and observations. No EDMS login, no separate system, no friction.

COMPLIANCE ARCHITECTURE

Regulation-ready by design.

Most QMS tools bolt compliance on as an afterthought. Vent is built for regulated environments from the first line of code — because it was designed by someone who lives those requirements.

21 CFR PART 11 / EU ANNEX 11

Electronic signatures with user ID and justification required on all approval transitions — QA sign-off, director approval, closure, rejection.

ROLE-BASED ACCESS CONTROL

Six distinct roles (Operator, QA, Director, MSAT, Engineering, Admin) with fine-grained route-level enforcement. Sensitive transitions locked by role.

IMMUTABLE AUDIT TRAIL

Every action — submission, status change, SOP modification, CAPA update, login — logged with SHA-256 checksums, timestamps, IP addresses, and user agents.

DATABASE-LEVEL SECURITY

Supabase Row Level Security policies. No direct table access without authentication. All queries proxy through the authenticated Express server.

ALCOA+ DATA INTEGRITY

Data is Attributable, Legible, Contemporaneous, Original, and Accurate. Anonymity for operators does not compromise traceability for the system — these are separated by design.

Why this matters for customers

FDA and HPRA inspectors audit electronic records systems. A platform without a compliant audit architecture cannot enter a GMP facility — full stop. Vent is inspection-ready from day one, which removes the single biggest procurement blocker.

MARKET OPPORTUNITY

A regulated, underserved, fast-growing market.

Biologics is the fastest-growing segment of pharma. Every facility runs on GMP. Every GMP facility has the same floor-to-QA information problem. None of them have a solution built for the floor.

TAM — Pharma Manufacturing Software **$18B**

Global market 2024 · 9% CAGR · 20,000+ GMP facilities worldwide

SAM — QMS, CAPA & Doc Control **$4B**

Biologics-specific quality management · ~5,000 target sites

SOM — AI-native Floor Intelligence (5yr) **$200M**

~500 sites · €35–60K ARR · expanding per-module

Ireland as the entry point

10+ major biologics MNC sites in the Cork/Dublin corridor (Pfizer, AbbVie, BMS, Regeneron, Eli Lilly, MSD). The world's largest pharma cluster, per capita. This is home territory.

BUSINESS MODEL

Site-based SaaS. Simple to land, strong to expand.

Annual per-site licences, with natural expansion across departments and up the management stack. CDMOs operating multiple sites are the highest-value enterprise accounts.

OPERATOR

€15K

per site / per year

- ✓ SOP Query (RAG-powered)
- ✓ Anonymous observation submission
- ✓ Document Builder
- ✓ Notification system

Entry point for floor teams

QA + WORKFLOW

€35K

per site / per year

- ✓ Everything in Operator tier
- ✓ QA Control Centre
- ✓ CAPA tracking & management
- ✓ SOP change management + audit trail
- ✓ Electronic signatures (21 CFR Part 11)
- ✓ 4-phase workflow routing

Core product — where value is captured

ENTERPRISE

€60K+

per site / per year

- ✓ Everything in QA tier
- ✓ Multi-site analytics & benchmarking
- ✓ Director intelligence dashboard
- ✓ API access + EDMS integration
- ✓ Custom SOP library ingestion
- ✓ Dedicated onboarding & validation support

CDMOs and multi-site operations

TRACTION & VALIDATION

Built, deployed, validated.

Pre-revenue. Post-prototype. The product exists, it works end-to-end, and the people it was built for have already asked when they can have it.

Live

Working product deployed on Railway. Full stack: submission → AI analysis → CAPA → QA workflow → director dashboard.

Solo

Designed, built, and deployed by a single founder — a practising bioprocess associate — while working full time. Zero external funding.

"Wizard"

Floor operator reaction on first product demo. Followed immediately by: "When can I actually use this?"

10+

Major biologics MNC sites within 2 hours of Dublin. All potential pilots. All running the same broken floor-to-QA loop.

What exists today

- ✓ End-to-end submission → AI analysis → CAPA workflow
- ✓ QA Control Centre with SOP cross-reference
- ✓ Document Builder with version control and AI Assist
- ✓ 21 CFR Part 11 compliant audit trail and e-signatures

What's next (with Dogpatch)

- → First paying pilot (CDMO or MNC, Ireland)
- → IQ/OQ/PQ validation package for GMP deployment
- → Visual SOP Engine (Phase 2)
- → Technical co-founder / first hire

PRODUCT ROADMAP

Phase 1 is done. This is where it goes.

The observation intelligence platform is the foundation. Each phase compounds the value of the last — more knowledge captured, more intelligence derived, harder to displace.

Phase 1 — Observation Intelligence Platform

DONE — 2026

Anonymous submission, AI SOP analysis, CAPA generation, QA workflow, document builder, audit trail, 21 CFR Part 11. Live at vent-production-8f8a.up.railway.app.

Phase 2 — Visual SOP Engine

Q3–Q4 2026

Operators capture photos and short videos during normal operations. AI maps media to SOP steps automatically. Builds rich visual walkthroughs for training. New starters watch the best operators perform every step — instead of reading 300-line text SOPs cold.

Phase 3 — Facility Intelligence Network

2027

Cross-facility pattern detection. Predictive deviation flagging. Operator expertise profiles and training competency mapping. Regulatory submission support and industry benchmarking across multiple sites.

Phase 4 — Industry Intelligence Platform

2027+

Aggregated, anonymised intelligence across the biologics industry. Facilities understand how their quality performance compares. Regulators see a platform that actively improves GMP compliance across the sector.

THE FOUNDER

Domain expertise is the moat.

DD

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Trinity College Dublin

> *"I am not a software person who discovered pharma. I am a pharma person who built software to solve a problem I live every day."*

FLOOR EXPERTISE

Works directly with operators on fed-batch and perfusion upstream processes. Knows their problems intimately — not from research, from daily experience.

CROSS-FUNCTIONAL

Works across QA, Manufacturing Support, and Engineering in cross-functional teams. Understands every stakeholder's position from the inside.

REGULATORY LITERACY

Understands the IQ/OQ/PQ validation requirements and audit trail standards needed to deploy software in a GMP environment. Vent is built to pass inspection.

LOOKING FOR

Technical co-founder (software/AI background). Advisory board with pharma commercial and regulatory expertise. Senior operators as design partners.

THE ASK

What we're asking for.

The product exists. The problem is real. The market is here. What I need is the platform to build this properly — and the right people in the room.

🏢

A place to build

A desk in Dogpatch so I can work on Vent full time — without splitting focus between the manufacturing floor and the laptop. This is the most important thing.

💶

Early-stage funding

A small seed grant (€25–50K) to cover infrastructure costs (AI APIs, hosting, database), free myself from employment, and reach a first paying pilot — without burning runway on anything else.

🤝

Network & mentorship

Access to Dogpatch's pharma and medtech connections in Ireland. Ireland is home to the world's densest biologics manufacturing cluster. An introduction to the right QA director or site head is worth months of cold outreach.

What you get

Early access to a founder who works inside the industry, knows every person on every floor, and has already built and deployed the product that solves a problem every GMP facility pays to manage badly. The moat is knowledge, and it compounds every day I stay on the floor.

- ✓ Live product you can demo today
- ✓ 10+ potential pilot sites within 2 hours
- ✓ Domain expertise no generalist founder can replicate
- ✓ A market that will only grow as AI enters regulated industries

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